


# Power Electronics

Report
Synchronous DC/DC Converter

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# Part I
# Introduction

The goal of this project is to design a simple bidirectional DC/DC power converter with some desired capabilities out of market bought components and simplified characteristics in order to showcase the workflow of completing such a task

# Part II
# Background

## Stages of design

The first stage of design of a power converter is to specify the working parameters of the system, those will be dependent on its application.
In this case the nominal values of the system were given. Using those the required component values can be calculated and matched from available products.
At this point the system can be simulated to see whether the system meets the specified parameters and whether the components needed require to be swapped out for more resilient ones. This can be done iteratively until the system behaves as expected.

## 1 Differences to an ideal design

There are a few factors that are taken into account in this projects. As the idea is to find stock components, the values found are exactly what is needed. These differences can have little impact such as higher maximum voltage and current allowed or could be more important ones such as the inductance values being much different for stock parts than what is actually needed, maybe even requiring commission of a custom coil from some manufacturer.
Another difference is the inclusion of internal losses for found components. In this case the conduction losses were only accounted for as other ones such as switching or thermal losses are beyond this course's scope. These while not significant will impact the required duty cycle of the system requiring manual tweaking or some kind of control system to avoid that. The internal resistance of the converter will also lose some power transfer efficiency a variable than can be tracked too. The last real design consideration that was ignored during this project was the inclusion of deadtime - a period of time before the complementary operation of the switches where both are open, which prevent the shorting of the circuit due to the both switches being functionally closed due to escape and recovery times of the components within.

# Part III
# Calculations Stage

Nominal Values:

- Nominal output voltage $\bar{V} = 750\,\mathrm{V}$
- Switching frequency $f_{sw} = 20\,\mathrm{kHz}$
- Current Ripple $\Delta i_L \leq 10\% \bar{i_L}$
- Input Voltage $V_{in} = 500\,\mathrm{V}$
- Input Power $P_{in} = 5.5\,\mathrm{kW}$

# Calculations for a Boost Converter

(1) $$i_L = i_{in} \implies P_{in} = V_{in} i_L \implies i_L = \frac{5.5\,\mathrm{kW}}{500\,\mathrm{V}} = 11\,\mathrm{A}$$

(2) $$v_c = v_{dc}\frac{1}{1-D} \implies D = 1 - \frac{v_{dc}}{v_c} = \frac{1}{3}$$

(3)
$$\Delta i_L \leq 10\% \bar{i_L} \implies \frac{|v_{dc} - v_c|}{L}(1-D)T \leq 10\% \bar{i_L} \implies L \geq |500-750|(1-\frac{1}{3})\frac{1}{20000}\frac{100}{11} \implies L \geq 0.00(75)[H]$$

(4) $$R_{on_{IGBT}} = \frac{2.6 - 2.3}{80 - 60} = 15\,\mathrm{m\Omega}$$

(5) $$R_{on_D} = \frac{2.5 - 1.8}{50 - 20} = 23.3\,\mathrm{m\Omega}$$

The average current through the converter Equation 1 is simply calculated as the ratio of the input power and input voltage using the nominal values, next the duty cycle for the ideal system is calculated in Equation 2. .Both of the previous values are used to find the inductance value Equation 3 to achieve the ripple was calculated too, the final chosen value of 7.6 mH should keep the system stable during buck operation when the current is inverted.The conduction resistances Equations (4) and (5) of the active components were calculated using the IGBT-diode package datasheet[1] as the slope of the linear part of their forward/collector-emitter current voltage characteristics. While a sufficiently resilient switch package was found, no stock inductor of the needed value capable of surviving the required current could be found. This means that the inductor needed for this converter would have to be made to order.

# Part IV
# Simulation

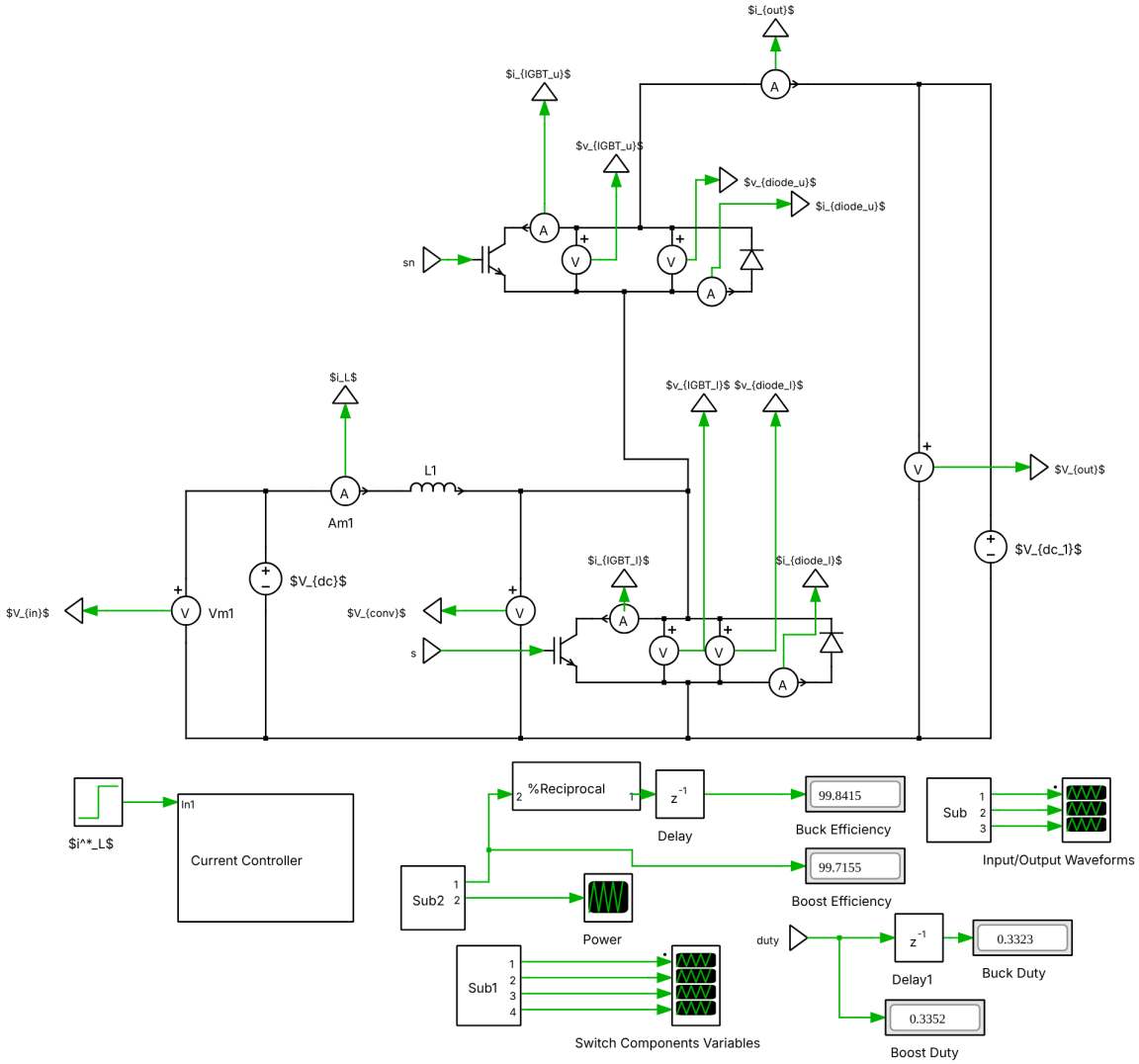



Figure 1: System Schematic

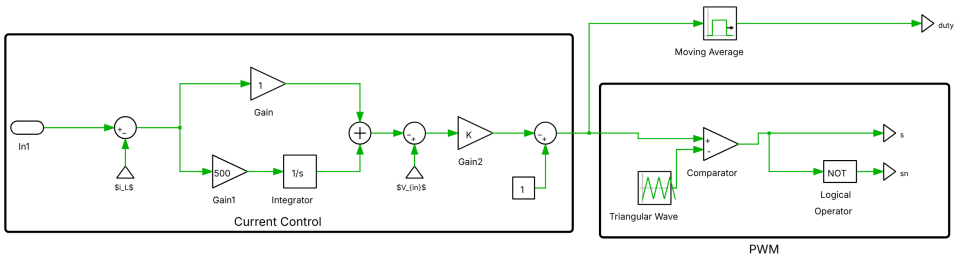



Figure 2: Current Controller

Once the system in Figure 1 has been assembled and fitted with a current controller in Figure 2 to avoid tedious manual tweaking of duty values to get the desired level of current the parameters of the system were collected.

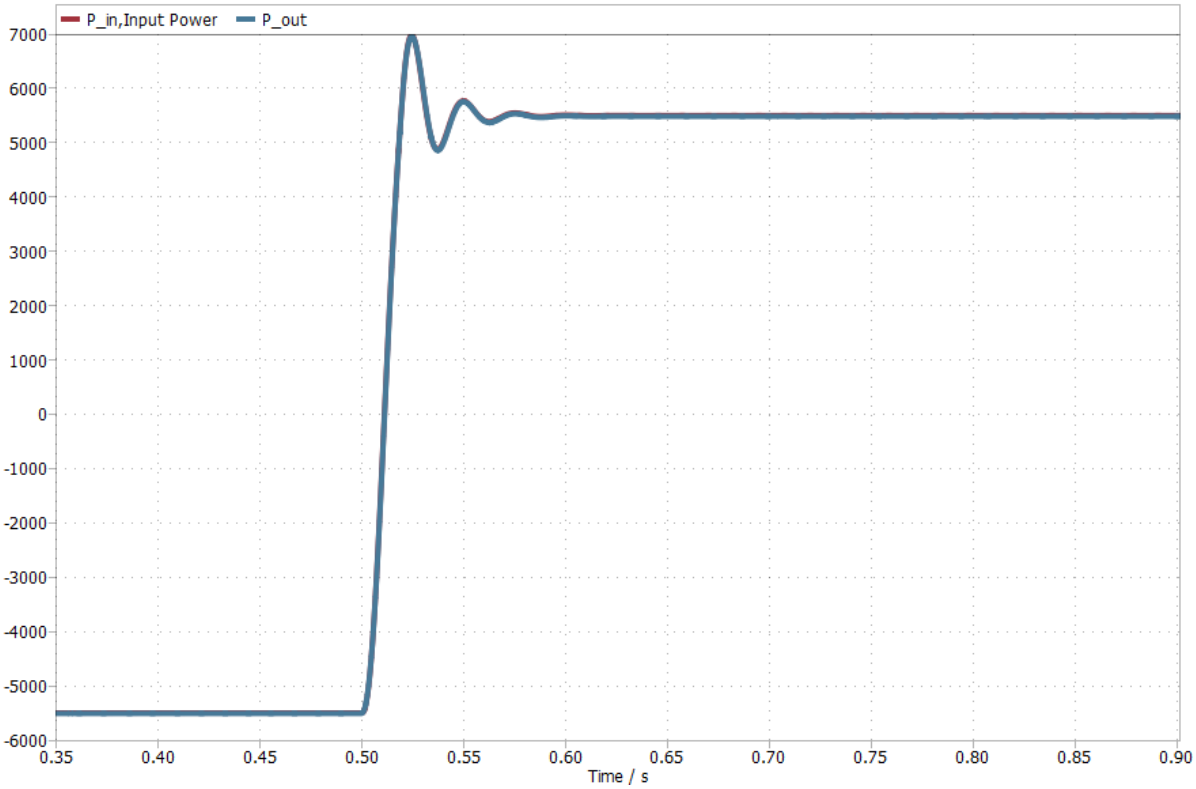
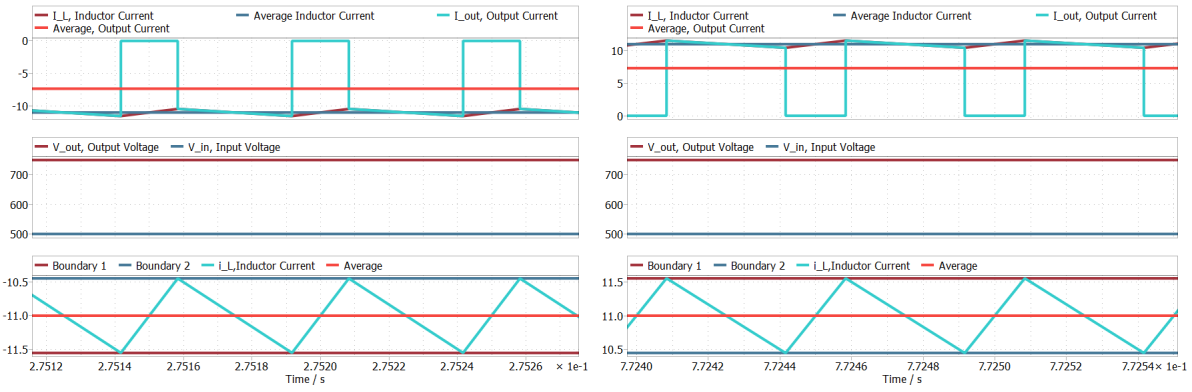



Figure 3: Power transferred through the system in the simulation

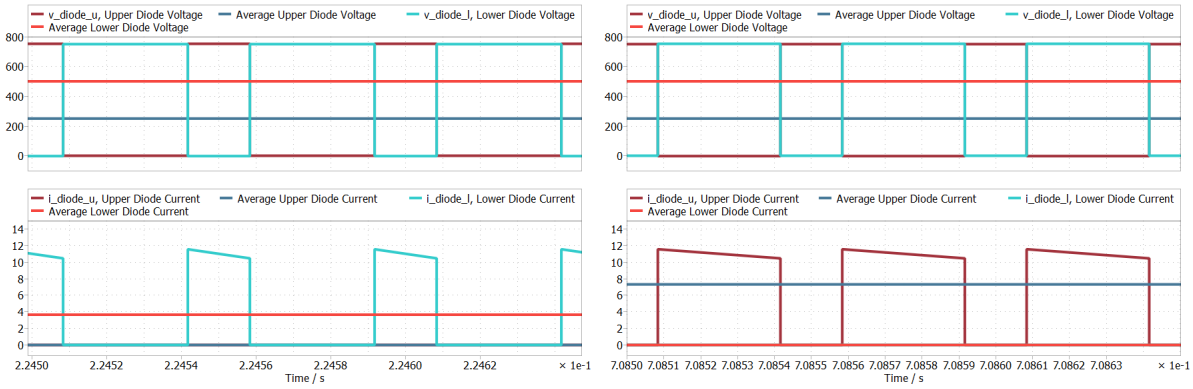



(a) Buck Operation

(b) Boost Operation

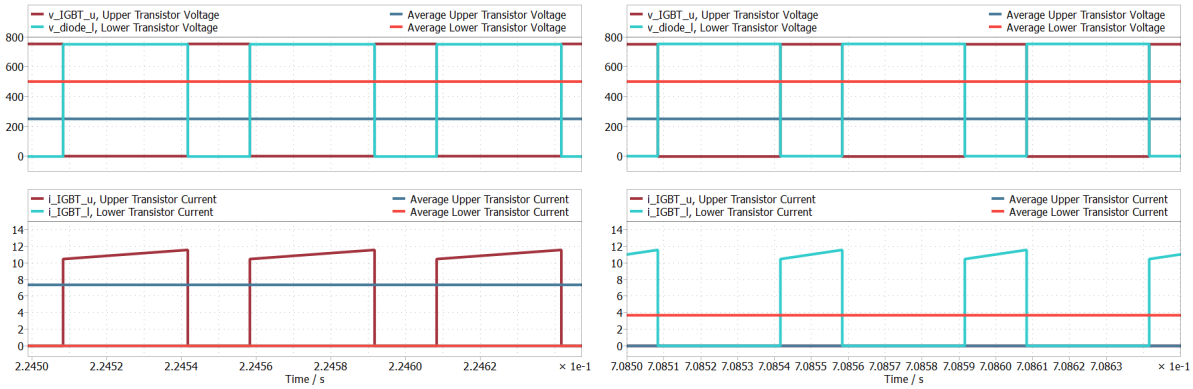
Figure 4: Input and Output Waveforms

As seen in Figures 3 and 4 the system is capable of bidirectional operation as it can transfer positive and negative power when current is inverted meaning it can operate both as a buck and boost converter. In both cases a steady state is achieved and nominal values are reached. In Figure 4 it can be seen that the ripple meets the specification too.




(a) Buck Operation (b) Boost Operation

Figure 5: Diode Voltage and Current




(a) Buck Operation (b) Boost Operation

Figure 6: IGBT Voltage and Current

By cross referencing the switch package datasheet[1] and Figures 5 and 6 it can be seen that the chosen component is adequate for this use case with generous headroom as the values seen are about half that of the maximum allowable ones. What is more is that commutation states of the converter can be observed here too as for buck operation it is the lower diode and the upper transistor which conduct current and opposite is the case for boost operation.

## Differences from ideal simulations

The main difference from the ideal simulations done previously on laboratories is that the efficiency of the system is not 100%, the values on displays in Figure 1 are very close too perfect though which is most likely due to the only the least impactful losses being implemented, nevertheless the maximum efficiencies of synchronous converters can be up to 98% so the point of efficiency of those is illustrated still. The efficiency of buck operation was higher than boost operation, which is due to the larger input current fraction going through the switch element with higher resistance. This difference has an additional effect on the system's stability as two different values of the duty cycle must be chosen for the usage of this converter as a buck or boost converter to compensate for those.

# Part V
# Conclusions

The converter designed should work from the electrical standpoint, though the workflow presented here would probably be conducted only as a preliminary analysis after which a more thorough simulation utilising spice programs where comprehensive behaviour of the components could be included. Nevertheless efficiency could be gauged and proper components selected based on it if thermal systems were to be designed separately.

## References

[1] https://www.infineon.com/dgdl/Infineon-IKW40N120H3-DataSheet-v01_10-EN.pdf?fileId=db3a304325305e6d012591d4832f7032. Accessed: 2025.04.11.